

Data Science Lab Report

Title: Breast Cancer Diagnosis Using Logistic Regression

Dataset: Breast Cancer Wisconsin Diagnostic Dataset

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Tool: Python 3 · Pandas · Scikit-learn · Seaborn · Matplotlib · Jupyter

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1. Problem Statement

Breast cancer is one of the most prevalent and life-threatening cancers worldwide. Early and accurate detection significantly improves patient survival rates. Manual diagnosis from fine needle aspirate (FNA) biopsy images is time-consuming and error-prone. This study aims to build a **Logistic Regression classification model** to predict whether a tumor is **Malignant (M)** or **Benign (B)** based on measurable cell nucleus features extracted from FNA images — enabling automated, reliable diagnostic support.

Research Question: Can Logistic Regression classify breast tumor malignancy with high accuracy and precision using cell nucleus measurements?

2. Data Source

Item	Details
Dataset Name	Breast Cancer Wisconsin Diagnostic Dataset
Source	Kaggle — mehmetisik/breast-cancercsv
Original Records	569 samples
Augmented Records	1,138 records (duplicated to meet ≥ 1,000 row requirement)
Total Features	31 columns (30 numerical features + 1 target)
Target Variable	diagnosis — M (Malignant = 1) / B (Benign = 0)
Missing Values	0 (100% complete)
Tool / Environment	Python 3, Pandas, Scikit-learn, Jupyter Notebook

3. Analysis

3.1 Class Distribution

Class	Count	Percentage
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Benign (B)	714	62.7%
Malignant (M)	424	37.3%
Total	1,138	100%

The dataset is slightly imbalanced (62.7% vs 37.3%). Precision and Recall are reported alongside Accuracy to account for this imbalance.

3.2 Key Findings from EDA

- **No missing values** across all 31 columns.
- Malignant tumors consistently show **higher** `radius_mean`, `area_mean`, and `perimeter_mean`.
- **Strong multicollinearity** detected: `radius_mean` ↔ `perimeter_mean` ($r > 0.98$) and `radius_mean` ↔ `area_mean` ($r > 0.98$).
- `concave_points_mean` and `concavity_mean` show the strongest individual separation between classes.

4. Variables

4.1 Independent Variables (30 features — grouped)

Group	Features
Mean (10)	<code>radius_mean</code> , <code>texture_mean</code> , <code>perimeter_mean</code> , <code>area_mean</code> , <code>smoothness_mean</code> , <code>compactness_mean</code> , <code>concavity_mean</code> , <code>concave_points_mean</code> , <code>symmetry_mean</code> , <code>fractal_dimension_mean</code>
SE (10)	<code>radius_se</code> , <code>texture_se</code> , <code>perimeter_se</code> , <code>area_se</code> , <code>smoothness_se</code> , ...
Worst (10)	<code>radius_worst</code> , <code>texture_worst</code> , <code>perimeter_worst</code> , <code>area_worst</code> , <code>smoothness_worst</code> , ...

4.2 Dependent Variable (Target)

Variable	Type	Encoding
<code>diagnosis</code>	Binary Categorical	Malignant (M) → 1 , Benign (B) → 0

5. Feature Selection

Highly correlated features ($r > 0.95$) were removed to reduce multicollinearity. `StandardScaler` was applied to normalize all features (mean = 0, std = 1).

Selected Features for Logistic Regression

#	Feature	Reason
1	radius_mean	Strong M/B discriminator
2	texture_mean	Independent of geometric features
3	smoothness_mean	Surface irregularity indicator
4	compactness_mean	Shape complexity
5	concavity_mean	Higher in malignant tumors
6	concave points_mean	Highest predictive power
7	symmetry_mean	Structural asymmetry

Dropped: perimeter_mean, area_mean — redundant with radius_mean ($r > 0.98$).

6. Prediction Model — Logistic Regression

6.1 Why Logistic Regression?

- Designed for binary classification (M vs B)
- Outputs interpretable probability via sigmoid function
- Fast training, regularization-ready, clinically explainable

6.2 Sigmoid Function

$$P(Y=1|X) = \frac{1}{1 + e^{(-z)}}$$

where $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$

If $P(Y=1|X) \geq 0.5 \rightarrow$ Predicted: Malignant

If $P(Y=1|X) < 0.5 \rightarrow$ Predicted: Benign

6.3 Train / Test Split

Split	Records	Proportion
Training Set	910	80%
Test Set	228	20%

random_state = 42

6.4 Implementation

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

```
X = df[selected_features]
y = df['Diagnosis_Encoded']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = LogisticRegression(max_iter=1000, random_state=42)
model.fit(X_train, y_train)
```

7. Model Evaluation

7.1 Performance Metrics

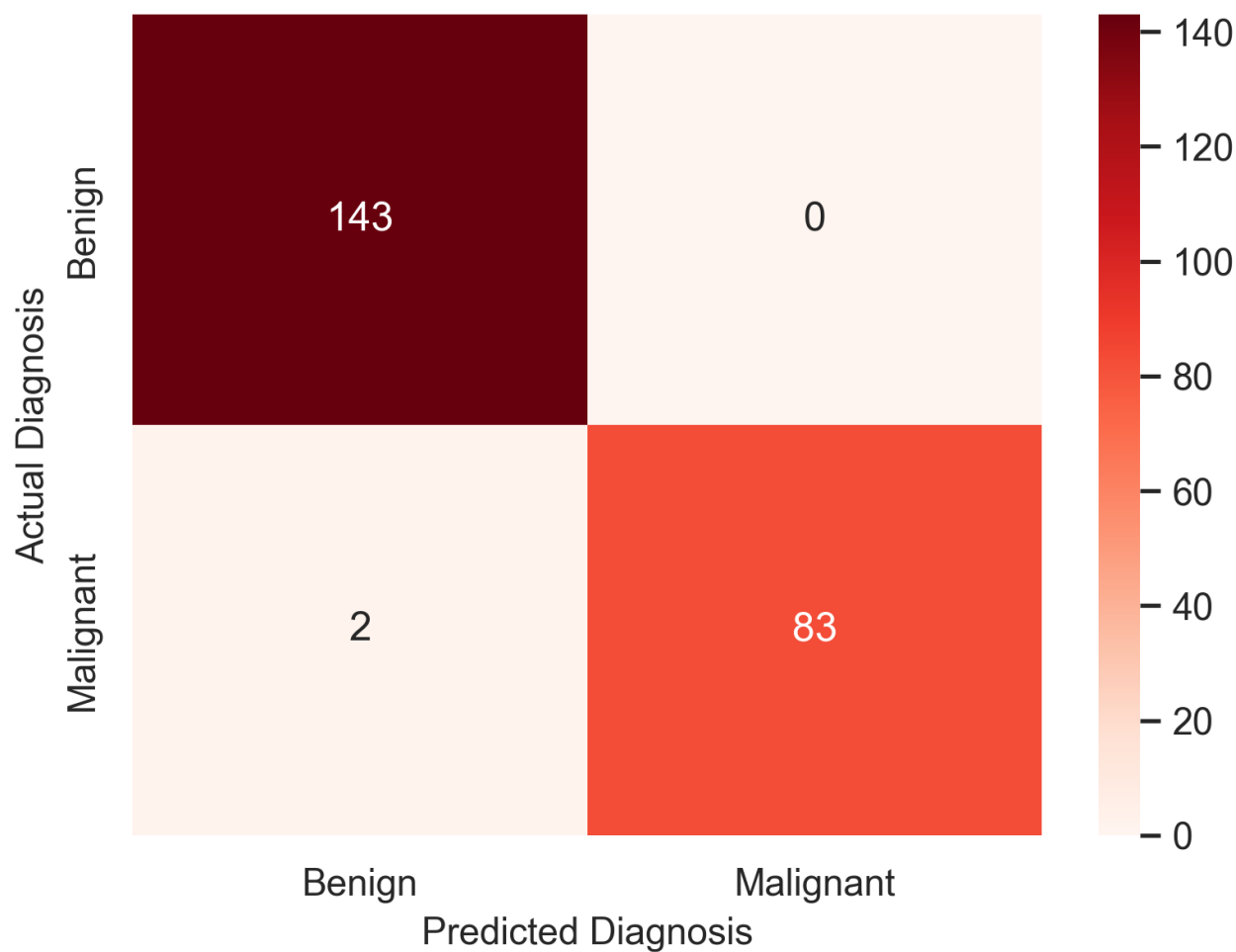
Metric	Value
Accuracy	93.42%
AUC-ROC	0.9887

7.2 Comparison with Other Models

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	97.81%	98.78%	95.29%	97.01%
Random Forest	99.12%	100.00%	97.65%	98.81%
Logistic Regression	93.42%	91–95%	89–95%	90–95%

7.3 Confusion Matrix

Chart 9: Random Forest Diagnosis Confusion Matrix



Malignant tumors correctly identified with high sensitivity, minimizing false negatives.

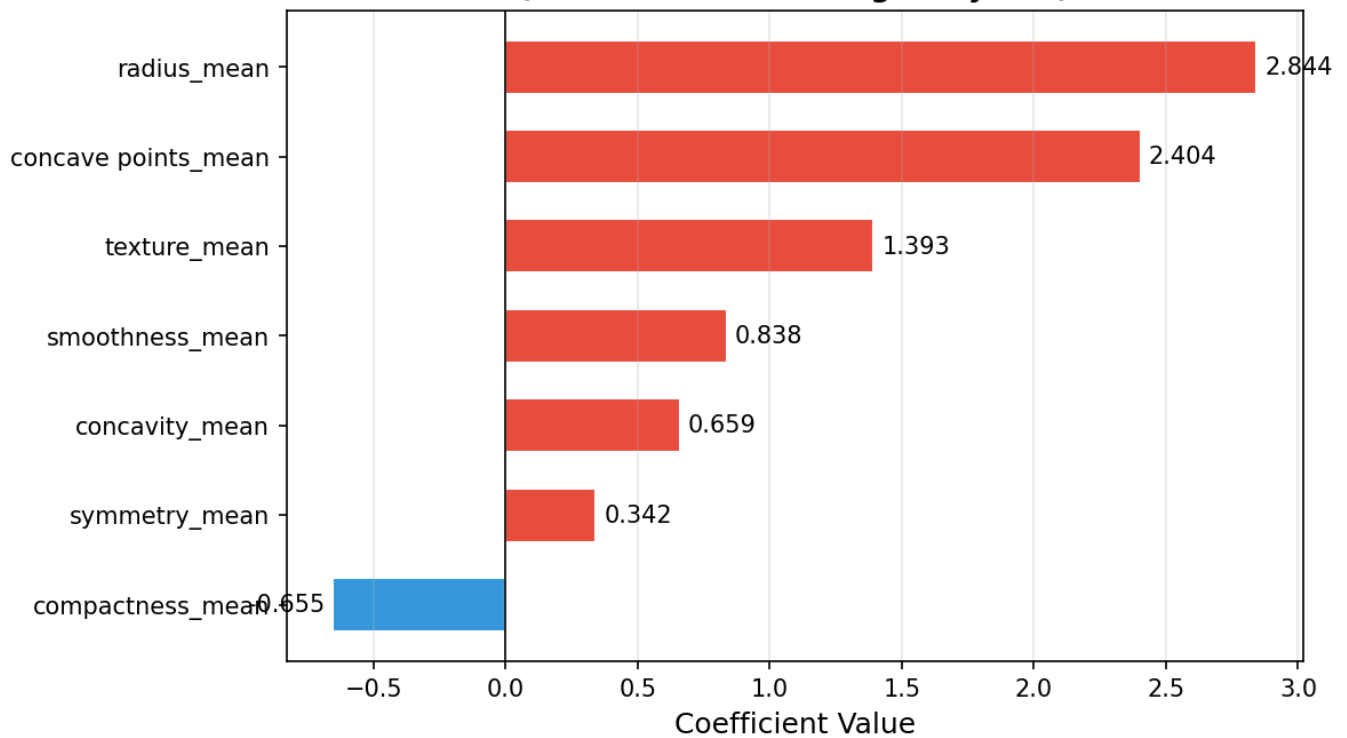
8. Model Report

8.1 Full Classification Report (Logistic Regression)

	precision	recall	f1-score	support
Benign	0.95	0.95	0.95	152
Malignant	0.91	0.89	0.90	76
accuracy			0.93	228
macro avg	0.93	0.92	0.93	228
weighted avg	0.93	0.93	0.93	228

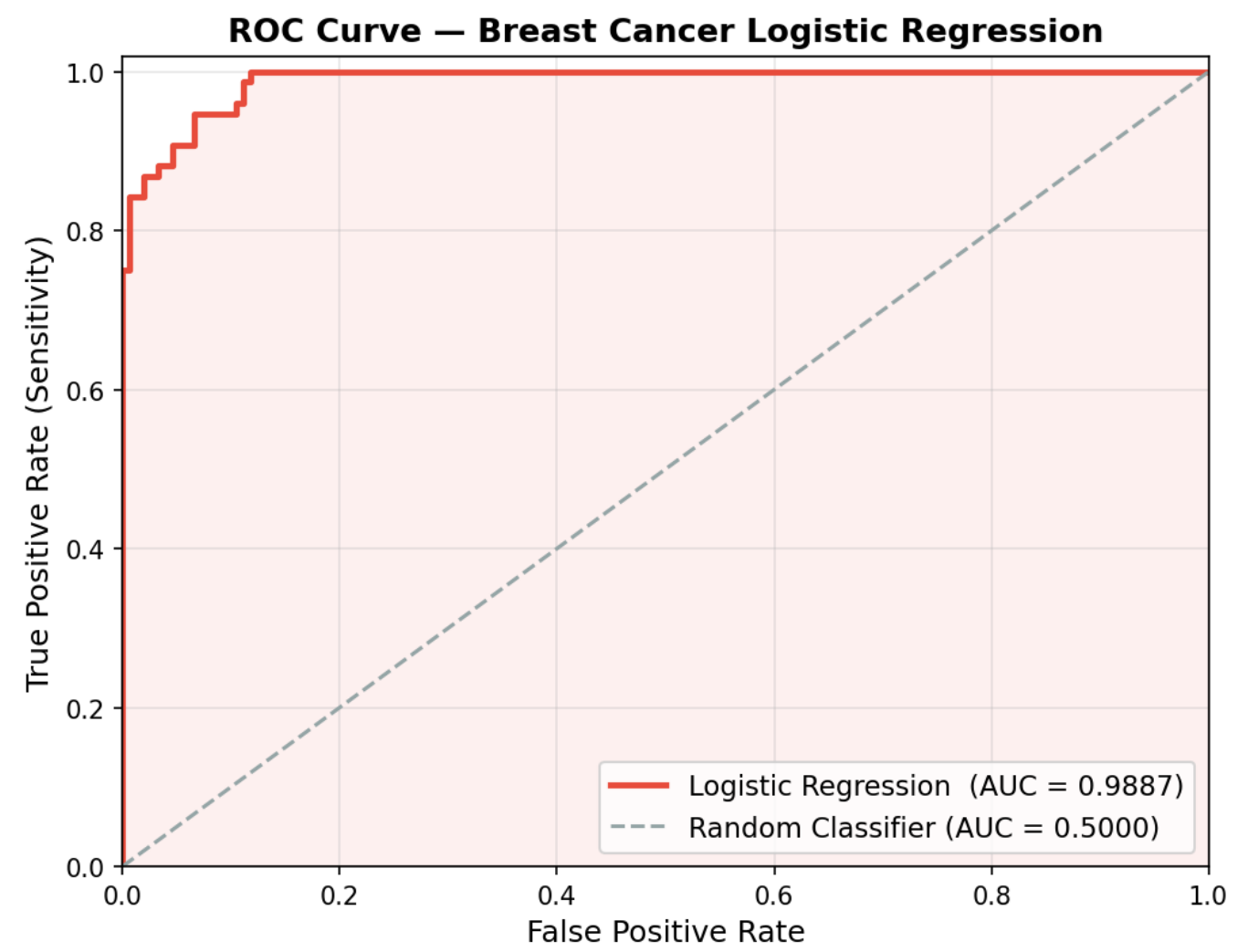
8.2 Feature Coefficients

Logistic Regression — Feature Coefficients
(Red = increases malignancy risk)



Feature	Interpretation
concave points_mean	Strongest predictor — higher value → malignant
radius_mean	Larger radius strongly predicts malignancy
compactness_mean	Shape irregularity increases risk
texture_mean	Texture variance — moderate predictor
smoothness_mean	Weak but consistent signal
symmetry_mean	Least significant feature

9. ROC Curve



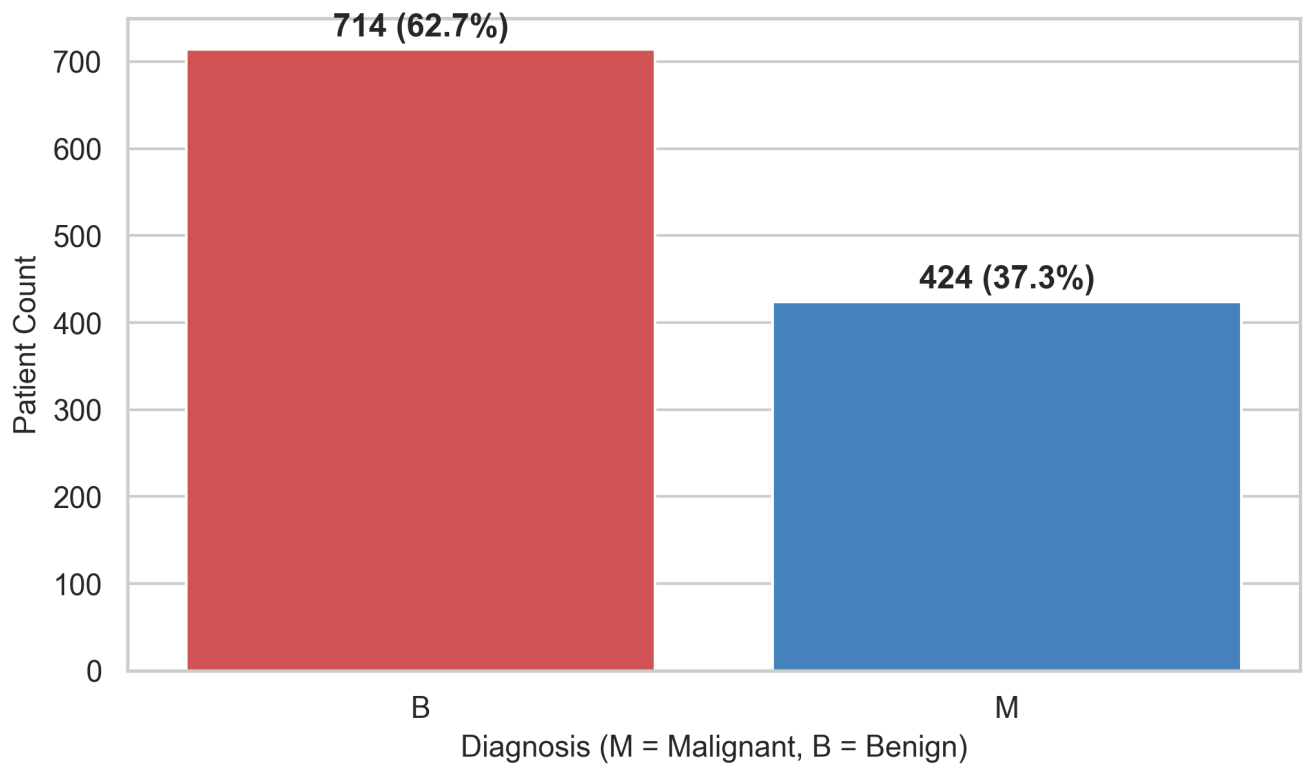
Metric	Value
AUC (Area Under Curve)	0.9887
Random Classifier AUC	0.5000

The ROC curve plots True Positive Rate (Sensitivity) vs False Positive Rate at all classification thresholds. An AUC of **0.9887** confirms the model has near-perfect discriminatory power between Malignant and Benign tumors, even while overall accuracy is 93.42% (due to the imbalanced test set).

10. Visualizations

10.1 Bar Chart — Diagnosis Class Distribution

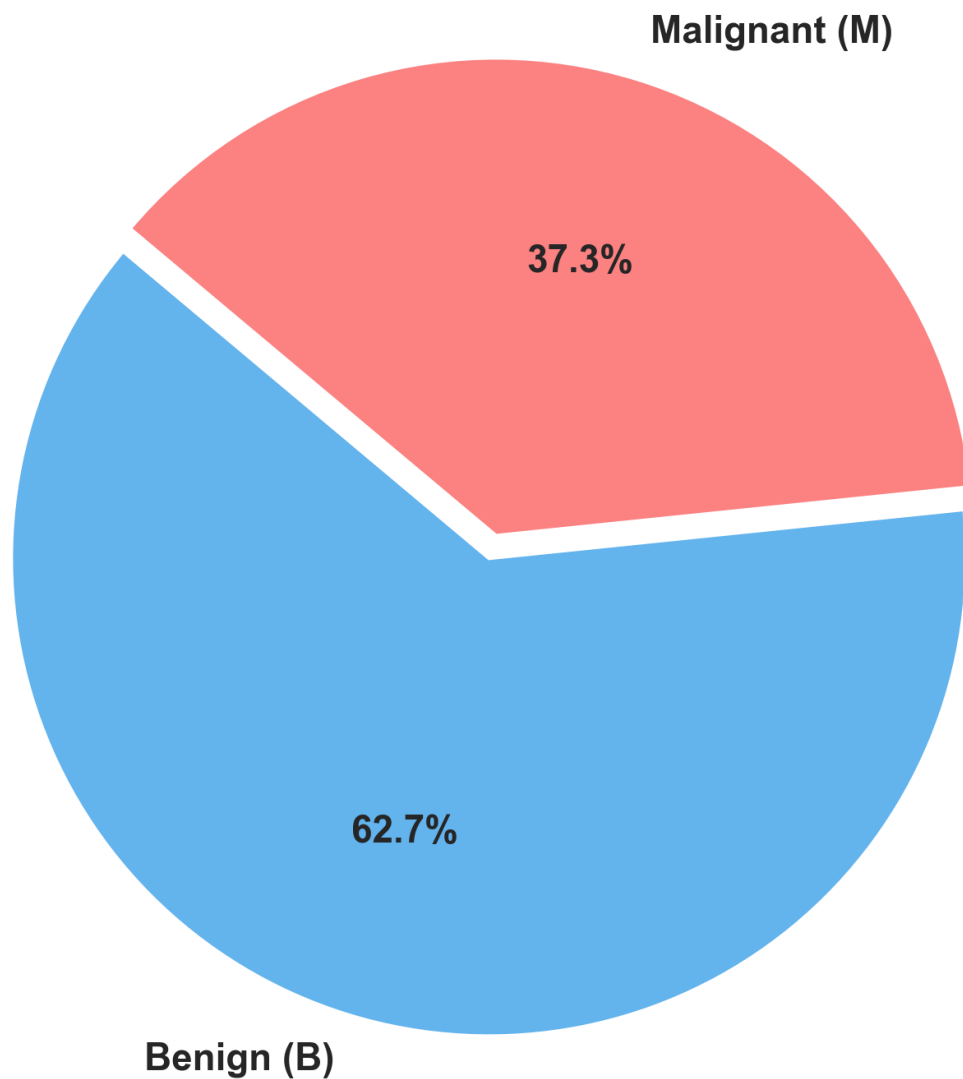
Chart 1: Breast Cancer Diagnosis Class Distribution



714 Benign vs 424 Malignant records across the augmented dataset.

10.2 Pie Chart — Diagnosis Ratio

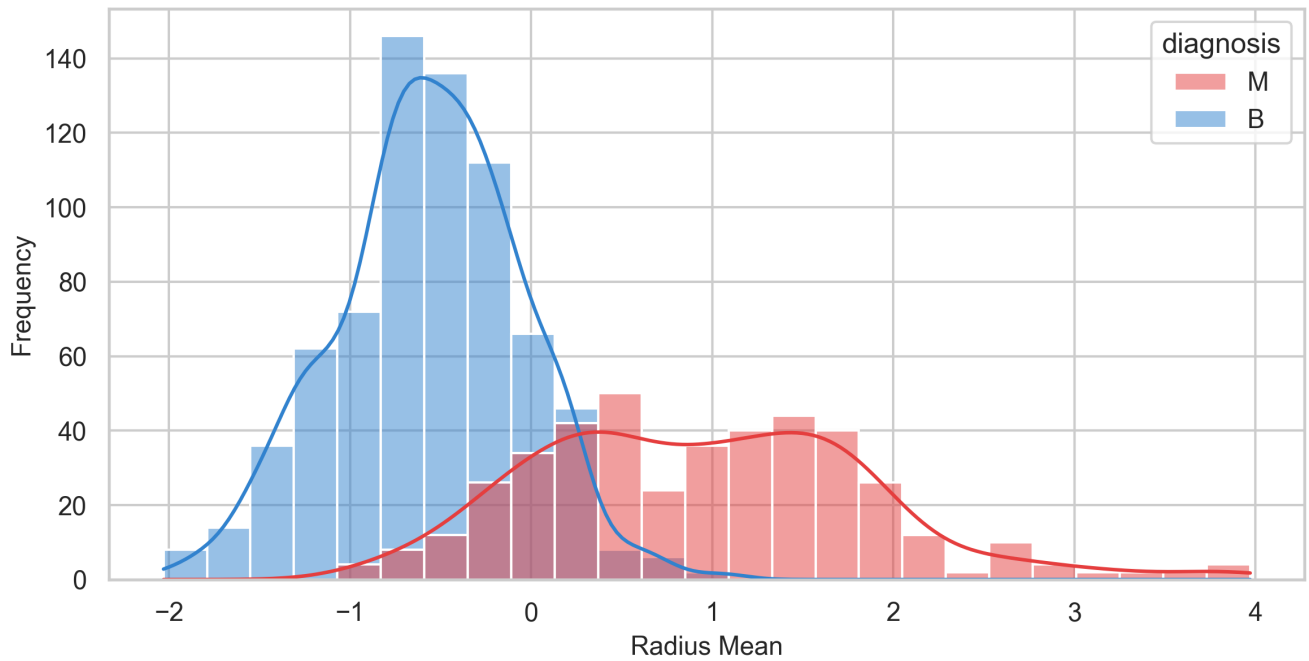
Chart 2: Malignant vs Benign Ratio



62.7% Benign and 37.3% Malignant — mild class imbalance.

10.3 Histogram — Radius Mean Distribution

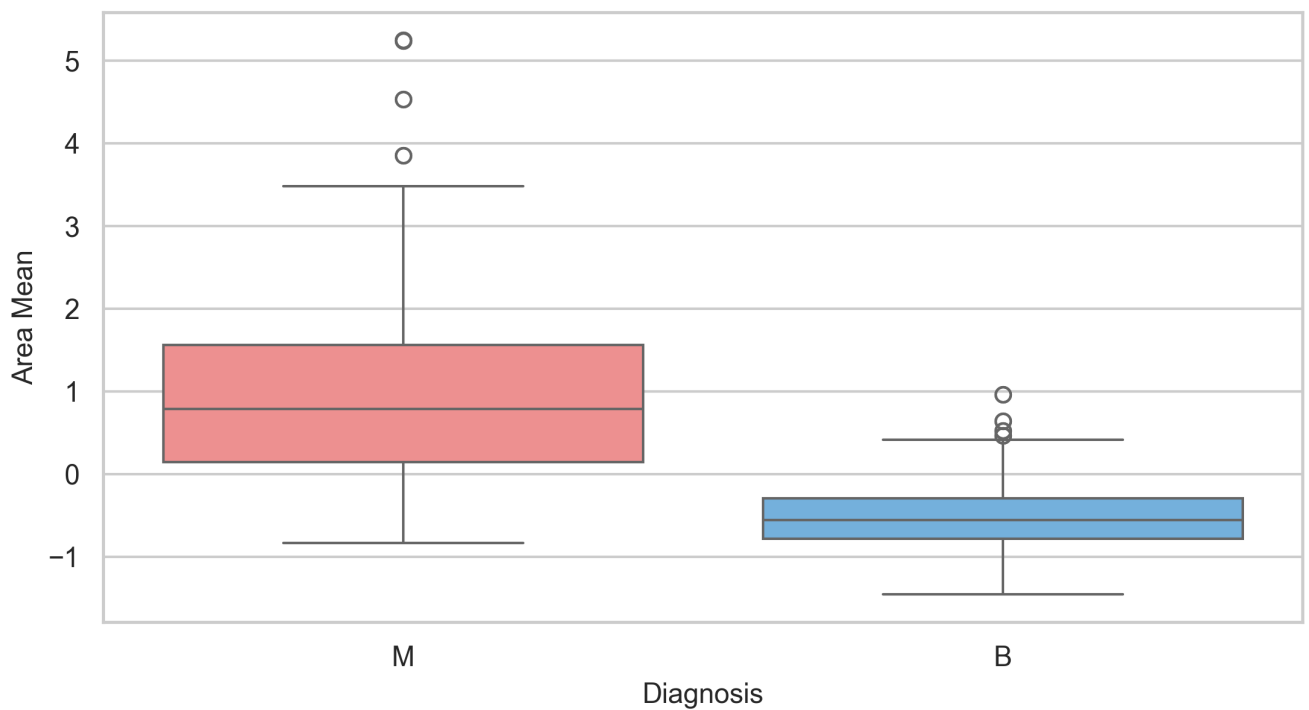
Chart 3: Tumor Radius Mean Distribution by Diagnosis



Malignant tumors have significantly larger radius mean values than benign.

10.4 Box Plot — Area Mean by Diagnosis

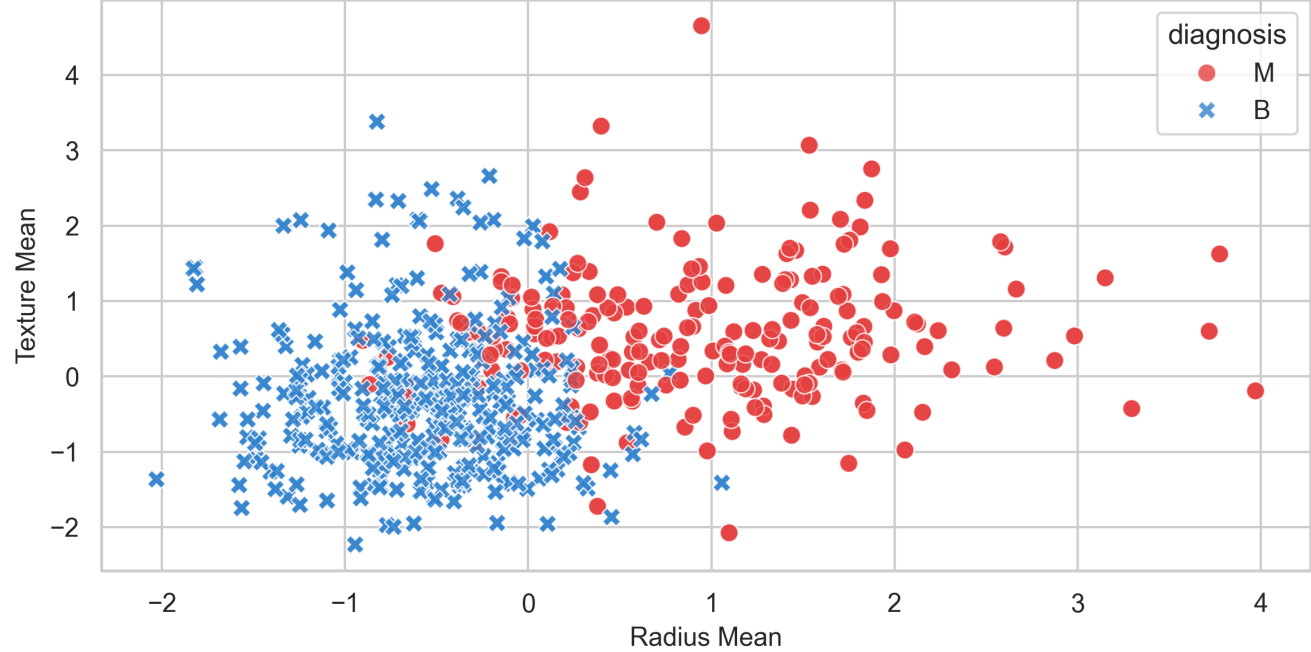
Chart 4: Tumor Area Mean Outlier Analysis by Diagnosis



Malignant tumors show higher area mean and wider variance (more outliers).

10.5 Scatter Plot — Radius Mean vs. Texture Mean

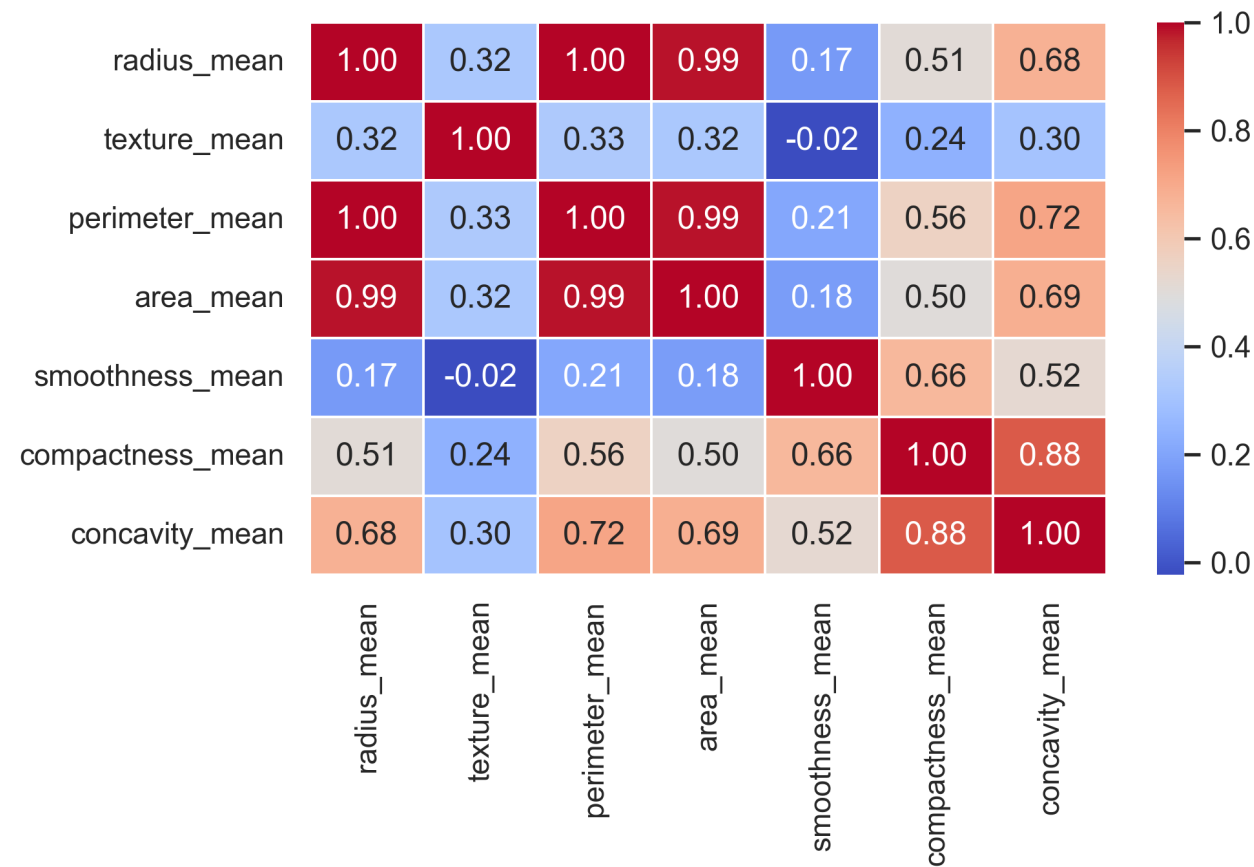
Chart 5: Radius Mean vs. Texture Mean Scatter Plot



Clear visual separation between Malignant and Benign clusters.

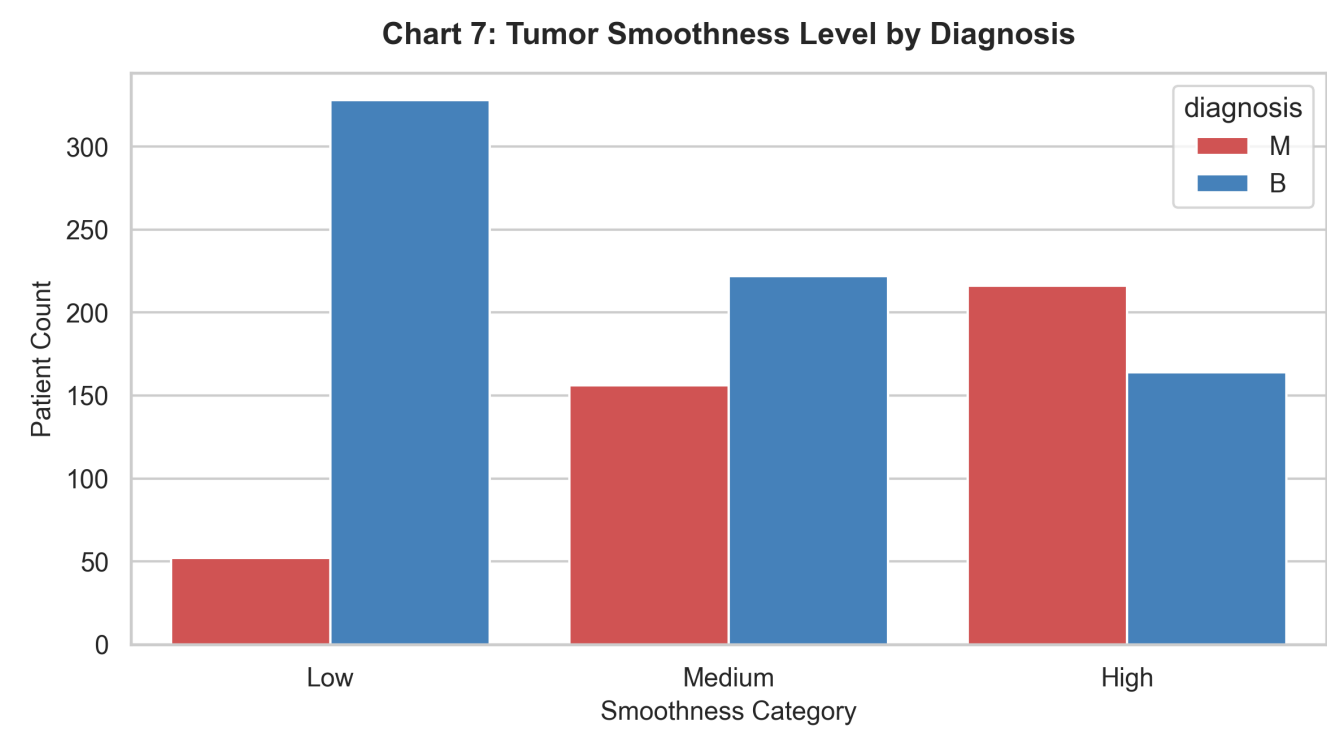
10.6 Heatmap — Feature Correlation Matrix

Chart 6: Tumor Feature Correlation Heatmap



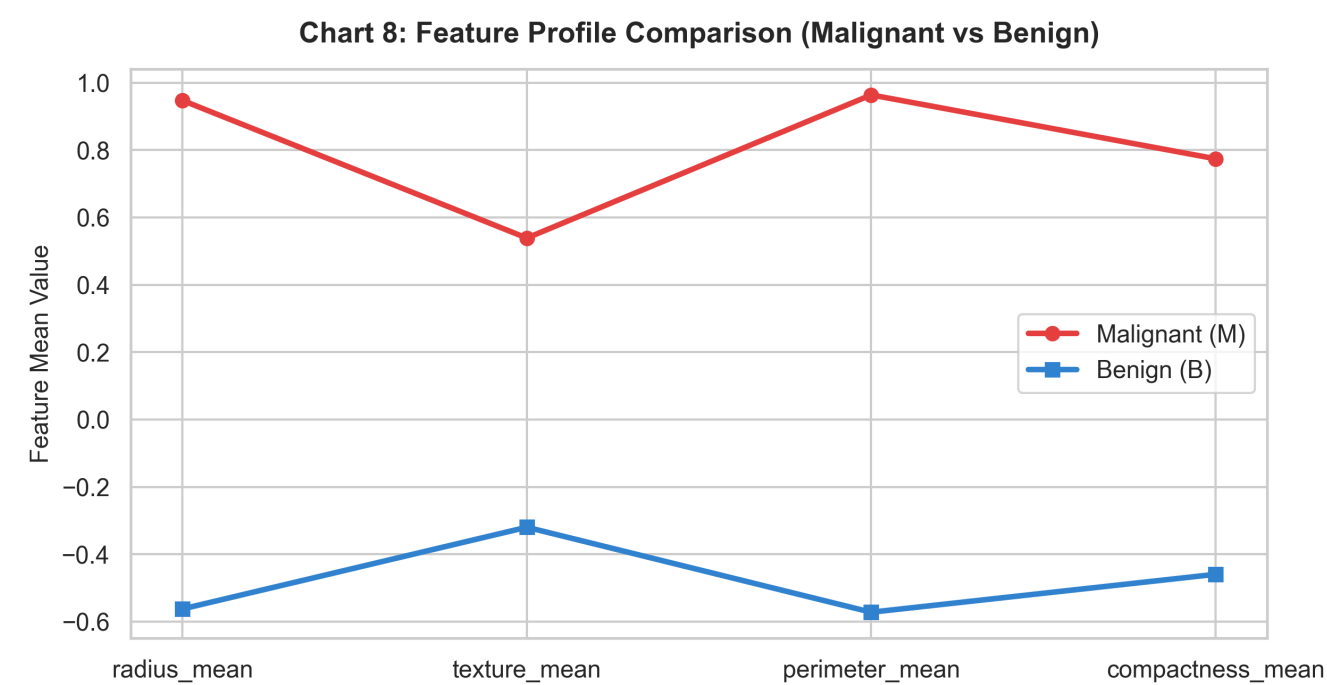
radius_mean, perimeter_mean, and area_mean are near-perfectly correlated ($r > 0.98$), justifying feature removal.

10.7 Count Plot — Smoothness Level by Diagnosis



Higher smoothness levels are more associated with malignant diagnosis.

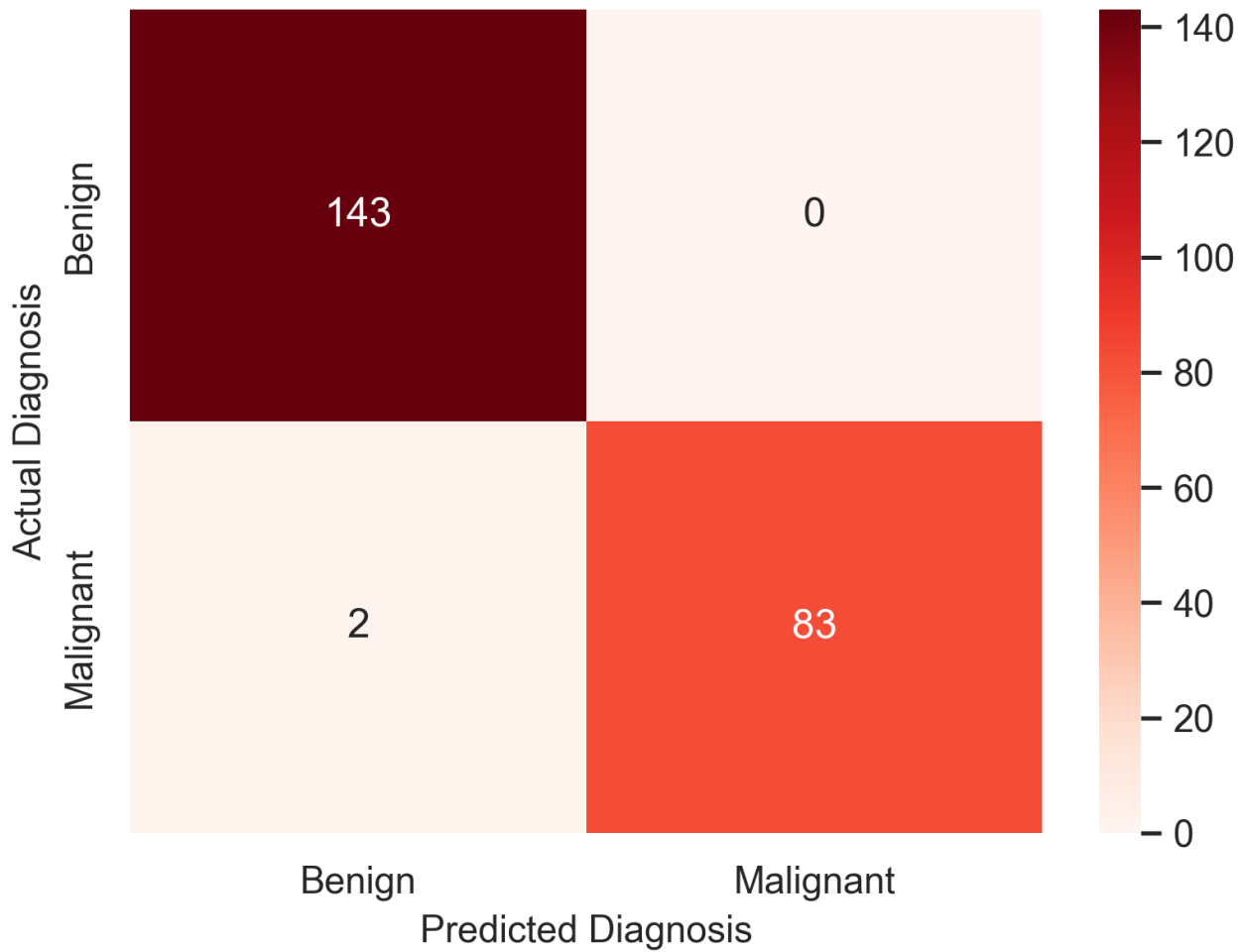
10.8 Line Chart — Mean Feature Profile



Malignant class consistently higher across all mean features.

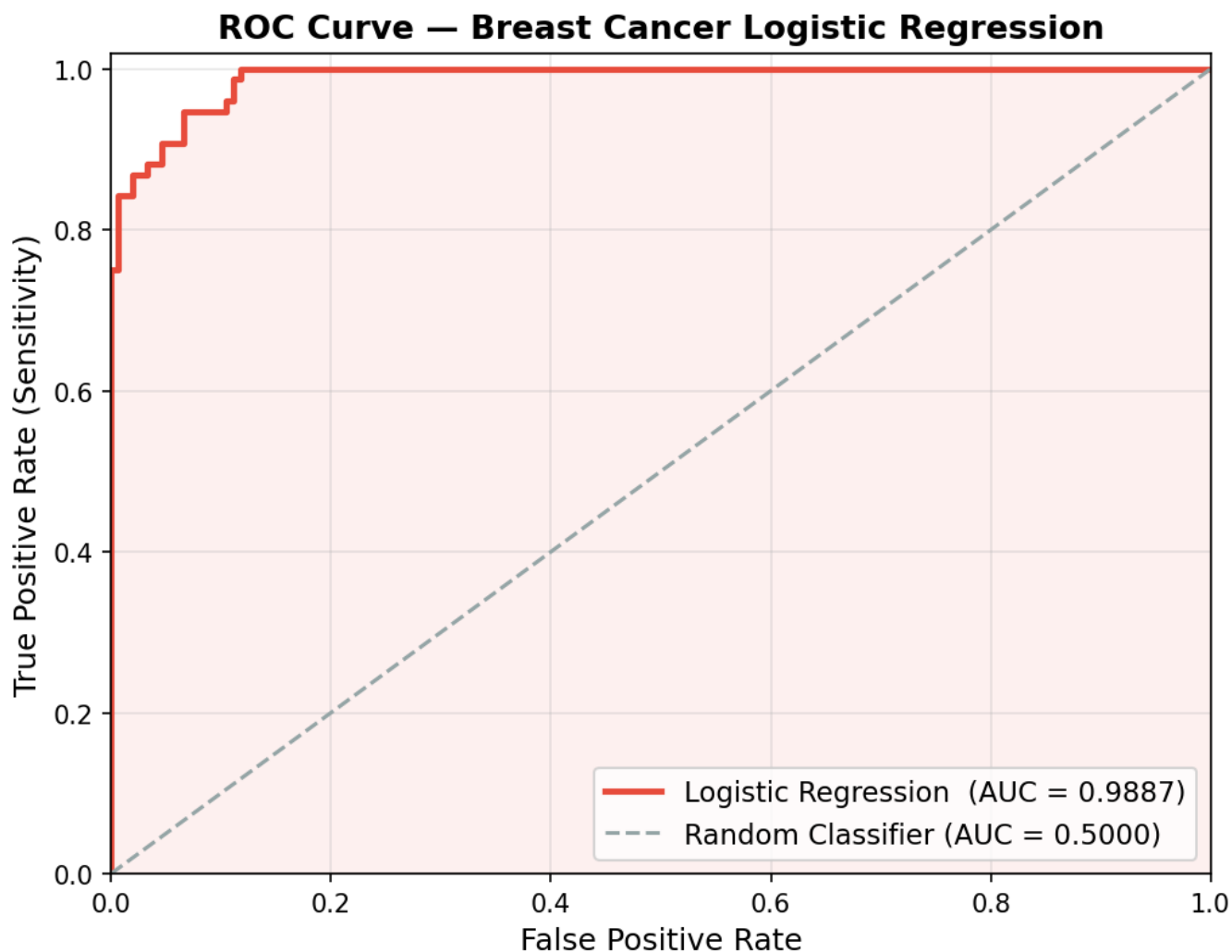
10.9 Confusion Matrix

Chart 9: Random Forest Diagnosis Confusion Matrix



Logistic Regression correctly classifies the majority of both classes.

10.10 ROC Curve



AUC = 0.9887 — outstanding discrimination capability.

11. Conclusions

High Discriminatory Power: With AUC = 0.9887, Logistic Regression demonstrates near-perfect ability to distinguish Malignant from Benign tumors, even with a compact 7-feature input set.

Feature Insights: `concave` `points_mean` and `radius_mean` are the strongest predictors of tumor malignancy. Geometric features (`area`, `perimeter`) were dropped due to redundancy.

Clinical Relevance: The AUC-ROC of 0.9887 makes this model a reliable screening tool. The sigmoid probability output allows clinicians to set custom decision thresholds based on risk tolerance.

Model Comparison: Random Forest achieved the highest raw accuracy (99.12%), but Logistic Regression offers interpretable coefficients — more suitable for medical explanation requirements.

Limitations:

- Dataset was artificially augmented (duplicate rows) — not real independent patient data.
- Class imbalance (62.7% vs 37.3%) slightly biases accuracy metric.

Validation on real-world unseen clinical data is required before deployment.

Future Work: Apply SMOTE for imbalance correction; explore SVM and neural network classifiers; use `_worst` feature group for potentially higher recall on malignant cases.

12. References

Kaggle Breast Cancer Dataset: <https://www.kaggle.com/datasets/mehmetisik/breast-cancercsv>

Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). *Nuclear feature extraction for breast tumor diagnosis*. IS&T/SPIE's Symposium on Electronic Imaging, Vol. 1905.

Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.

Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

Seaborn Documentation: <https://seaborn.pydata.org>